3.RecyclerView

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Student List"
        android:textSize="25sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="10dp" />

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

Create item_student.xml
Right-click **res → layout → New → Layout Resource File**

item_student.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="15dp">

    <TextView
        android:id="@+id/txtName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/txtCourse"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:textSize="16sp" />

</LinearLayout>
```

Create StudentAdapter.java
Right-click your package → **New** → **Java Class**

StudentAdapter

```
package com.example.recyclerviewdemo;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;


import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

public class StudentAdapter
      extends RecyclerView.Adapter<StudentAdapter.StudentViewHolder> {

   String[] names = {
         "Amit",
         "Rahul",
         "Sneha",
         "Pooja",
         "Rohit"
   };

   String[] courses = {
         "BSc IT",
         "BSc CS",
         "BCA",
         "MCA",
         "BSc IT"
   };

   @NonNull
   @Override
   public StudentViewHolder onCreateViewHolder(
         @NonNull ViewGroup parent, int viewType) {

      View view = LayoutInflater.from(parent.getContext())
            .inflate(R.layout.item_student, parent, false);

      return new StudentViewHolder(view);
   }

   @Override
   public void onBindViewHolder(
```

```
        @NonNull StudentViewHolder holder, int position) {

      holder.txtName.setText(names[position]);
      holder.txtCourse.setText(courses[position]);
   }

   @Override
   public int getItemCount() {
      return names.length;
   }

   static class StudentViewHolder extends RecyclerView.ViewHolder {

      TextView txtName, txtCourse;

      public StudentViewHolder(@NonNull View itemView) {
         super(itemView);

         txtName = itemView.findViewById(R.id.txtName);
         txtCourse = itemView.findViewById(R.id.txtCourse);
      }
   }
}
```

MainActivity.java

```
package com.example.recyclerviewdemo;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
public class MainActivity extends AppCompatActivity {

   RecyclerView recyclerView;

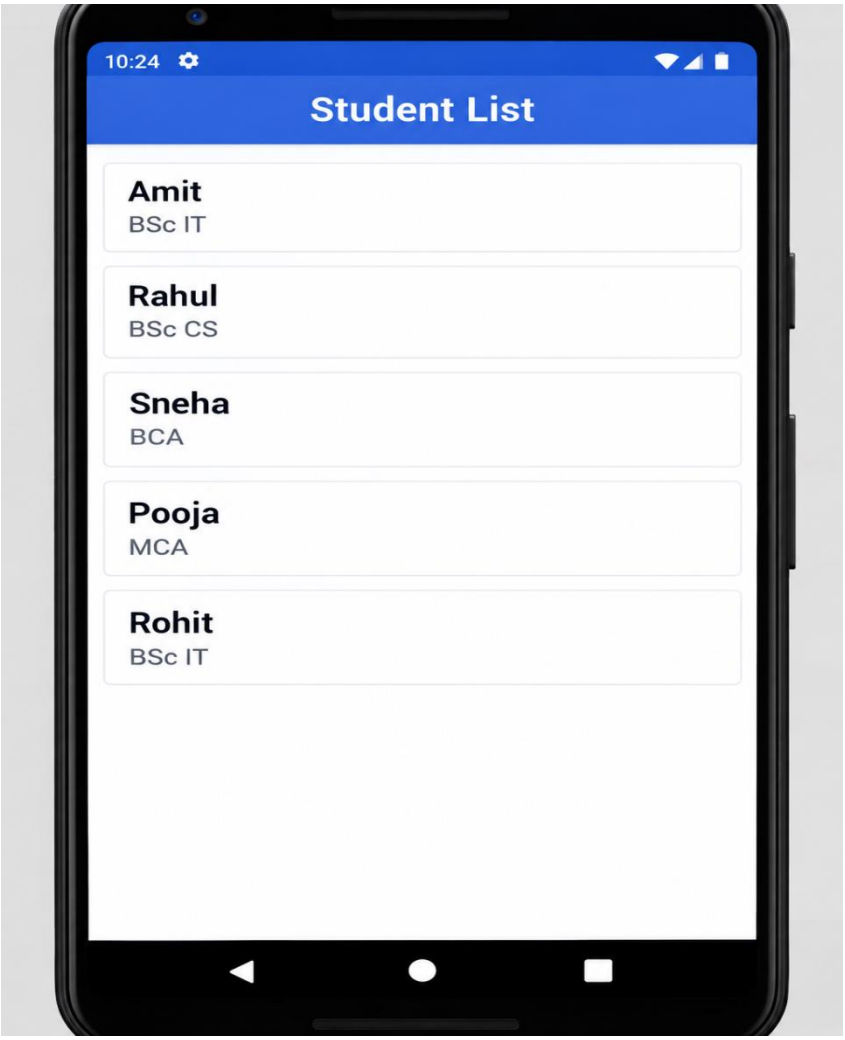
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      recyclerView = findViewById(R.id.recyclerView);

      recyclerView.setLayoutManager(
            new LinearLayoutManager(this));

      StudentAdapter adapter = new StudentAdapter();

      recyclerView.setAdapter(adapter);
   }
}
```


10:24
Student List
Amit
BSc IT
Rahul
BSc CS
Sneha
BCA
Pooja
MCA
Rohit
BSc IT